

Show or Tell?

A Comparison of Direct Instruction Tutorial and Learn by Doing Increased Impasse Versions of Initial Levels of a Puzzle Game

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ABSTRACT

Game designers often utilize tutorials to explicitly introduce players to game mechanics. Another design approach is to begin by throwing players into the “shallow end”, introducing them to mechanics through active engagement in easier levels. To compare these different approaches, we created two modified versions of the first 7 levels of the puzzle game *Baba Is You*: a tutorialized version, and an increased impasse version. Seventy-five players spent up to 20 minutes playing one of the modified versions or the original game and were asked to fill out a survey on their engagement level, perceived difficulty, and understanding of the core game mechanics. Analyses show that players of the tutorialized version made fewer mistakes and rated their version at a lower difficulty than players of the original and increased impasse versions, while all groups scored highly on content knowledge and engagement. This suggests that direct instruction at the start of a game allows players to come to an understanding equal to a learn-by-doing approach more quickly, with fewer mistakes, and without sacrificing engagement. This study does not, however, show that tutorial players are more prepared to take on more challenging content without explicit guidance, a direction proposed for subsequent work.

CCS CONCEPTS

• **Applied computing** → Computers in other domains; Personal computers and PC applications; Computer games; Education; Interactive learning environments.

KEYWORDS

Tutorial, Design Choice, On-boarding, Learning

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1 INTRODUCTION

One of the most pivotal choices a game designer makes is how to teach the player the basic mechanics. Tutorial areas are a common approach game designers use to prepare players to navigate their game [1–4]. These areas often come at the start of the game and are designed to show the player how the basic mechanics work. Typically, this is through just-in-time [5] pop-ups detailing how the controls work and what the player must do to progress. Akin to direct instruction [6], this method of learning is also often seen in educational contexts, including lectures and textbooks. Another common strategy, especially in puzzle games, is to throw the player into the “shallow end,” teaching them how to play via simple puzzles that require the player to discover and use the basic mechanics, demonstrating how they work and giving them tangible experience using them. Both of these strategies align with learning principles that game scholars argue encourage players to learn the basic mechanics of the game and progress through more difficult content [5–8]. However, it is unclear if either approach, explicit instruction or learn-by-doing, teaches players about the mechanics more efficiently.

In this study, we created two versions of the first 7 levels of the popular puzzle game *Baba Is You* [9]. One version was modified to create a more explicit instruction tutorial, wherein players are instructed how to complete each section of the level, visually guided to objects of importance, and given reviews of important mechanics upon level completion. The other version is modified to increase the likelihood that players will encounter impasses that require the player to think about how the mechanics of the game work. We gauge player engagement, perceived difficulty, patterns in impasse encounters, and player’s knowledge of the core game mechanics. Results show that players of both versions of the game learn the mechanics equally well and report the same levels of engagement as players of the original game. Players of the tutorialized version of

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the game report lower levels of difficulty and show fewer gameplay behaviors that suggest they are stuck or made a mistake. This suggests that these disparate strategies of teaching players the mechanics of a game only differ in the perceived level of difficulty and not the level of understanding imparted to players.

2 RELATED WORK

Puzzle games are a prolific and broad genre in the video game industry, ranging from small indie titles based on solitary yet clever concepts that create a collection of puzzles, to large design company titles that include deep narratives in expansive worlds that tie puzzles together as players progress [10]. While some games include puzzles but are not considered part of the puzzle game genre, researchers have defined puzzle games as “video games where problem-solving is the core and main mechanic and where the main source of satisfaction is solving these problems” and where the player is expected to “focus on pattern recognition, matching and creation. They require players to concentrate, to think about their moves, to move towards solving a challenge that they accepted. A puzzle is, in the end, a problem with a non-obvious solution” [11]. As such, puzzle games need to teach players how the mechanics of the puzzles work well enough to be able to figure out their solution autonomously. This requires walking a fine line between instructing players enough that they can progress and instructing too much that the puzzle’s solution becomes trivial. In working through the complexity of teaching players the mechanics of a game, we focus on two common approaches.

2.1 Direct Instruction Tutorials

Tutorials are often hard-wired into the early parts of a game [1], and give designers a chance to explicitly impart to players the game’s foundational mechanics. Without effective instructions, players may find themselves frustrated and unsure how to progress. A wide array of design strategies are used for tutorials, some with only instructions about the controls available to the player and not how any elements of the game interact, requiring the player to discover these themselves. With the increasing complexity of modern games, tutorials have become even more of a key part of bringing the player into the game world [1, 2]. Creating an effective tutorial is not trivial. In prior research on the structure of these tutorials [4], several dimensions are considered relevant, including when, where, and how to present a tutorial. Most often, tutorials use a combination of written or spoken instructions, demonstrations, and experiences at the start of a game [2]. Through a learning science lens, this approach is akin to direct instruction, in which learners are taught an explicit strategy before being encouraged to develop their mastery by correcting mistakes as they practice a wide range of examples [6]. While this approach has been widely used in classrooms, other more learn-by-doing approaches are also used in educational settings as well as in early levels of games.

2.2 Shallow-end Approach

Another common design strategy to ensure players learn the core mechanics of a game is through a game design technique that throws them into the “shallow end”. This is done through simple puzzles at the start of the game that require the player to use

the core mechanics, referred to by game designer Mark Brown as an Invisible Tutorial [12]. By grappling with mechanics early, players can develop some understanding of how they work. This approach is common in puzzle games as it ensures that the solution to these early puzzles is not spoiled or feels trivial. Also, part of “onboarding,” [13, 14] this approach intentionally uses easier but still challenging sections of the game to give players time to understand the basic mechanics before they move on to more difficult parts of the game. Likewise, this approach can also be viewed through a learning science lens as akin to constructionism, in which learners are encouraged to “learn by doing,” encountering and evaluating impasses to understand better the system in which they are working [7, 8]. Put differently, by actively engaging with the problem space, learners have the opportunity to “bump up against the world to test their thinking” [15]. By testing their ideas, learners can identify key features of the problem and *construct* a coherent mental representation of the targeted concepts. This approach has also led to a branch of research that details “productive failure”, in which learners are encouraged to “fail” while initially navigating a problem space, resulting in a better understanding of the material once scaffolded by an expert [16, 17]. In games, this is seen through these early levels that encourage players to learn the core mechanics by interacting with them, evaluating any impasses they encounter to better understand how the mechanics work through concepts like well-ordered problems and fishbowls [5, 18].

2.3 Current Work

Previous work comparing the presence or absence of tutorials in the learnability of games shows that tutorials are effective in more complex games, but do not necessarily increase engagement and retention in simpler games [1]. This work presents relatively basic tutorials in three different games, and finds that for simpler games, learning through exploration is as effective as presenting instructions, without the time-consuming task of creating and presenting a tutorial. While this suggests that tutorials are key in certain complicated games, an instructional-design question remains around how to introduce complex mechanics during this tutorial period – explicitly, with direct instructions, or implicitly, by designing to encourage learn-by-doing.

In the context of a complex puzzle game, *Baba is You*, we expand this work by comparing three game versions: a tutorialized version that provides direct instructions, a version that increases impasses, and the original, unaltered version of the game. We explore how each approach impacts player performance, understanding of the base mechanics, problem-solving abilities, and engagement. Through this work, we aim to answer the following question: *How do direct instruction tutorials and learn-by-doing impasse-focused approaches to learning in the initial levels of a puzzle game differ in generating understanding of the basic mechanics and engagement in players?*

3 METHODS

To compare the efficacy of tutorial-based levels to increased impasse levels at the start of a puzzle game, we created two versions of the initial 7 levels of the popular puzzle game, *Baba Is You*. In the original version, these levels present players with puzzles that



Figure 1: : Level 1 Original Game

encourage them to tinker with word tiles that create rules governing the physics of the level. For example, when level 1 loads, the player finds that when they use the movement controls, they are currently controlling the “Baba” figure and are surrounded by walls and three-word tiles in a row that say “WALL” “IS” “STOP” (Figure 1). If the player explores, they find that the walls are impassable until they disrupt the word tiles that form the rule “WALL IS STOP”, after which they can walk through the walls because they are no longer a barrier. Once in the top area they find there is no object they can touch to win and they need to form the rule “FLAG IS WIN”. Through the first 7 levels, the players learn about other mechanics like “Stop” or “Win”, that any object can be “You”, that “Sink” causes objects to disappear along with the thing they sink into, that “Hot” objects disappear if they touch an object that is “Melt”, and many other mechanics that build upon each other to create unique learning opportunities.

3.1 Tutorial Version

To create a tutorialized version of the initial 7 levels, we started with the layout of the original puzzles and added features that would explicitly guide the player through each level section, explaining what the mechanics do and how they need to use them to progress. For this version, level one loads to the same layout, but within

the walls is a signpost that informs the player that they need to “Break the WALL IS STOP rule to move through the walls!” and presents visual cues that point their attention towards the game objects they need to manipulate (Figure 2). Once the player follows these instructions, they gain access to the next area which contains another signpost that informs them they need to “Move the words together to make FLAG IS WIN and touch the flag!”. Once they follow these instructions and complete the level, they are then met with another signpost that reinforces the core mechanics of the level - “In this level we learned about STOP, breaking rules, and making a win condition!” Through the rest of the initial 7 levels, the player is guided by signposts that instruct them how to use the relevant mechanics to progress, direct their attention to objects of importance, and recap the important mechanics of each level.

To create an increased impasse version of the initial 7 levels, we started with the layout of the original puzzles and added features to increase the likelihood players would physically encounter impasses relevant to the mechanics needed to progress through each section of each level. For example, level one loads with a slightly different layout than the original. Instead of being surrounded by walls, players find themselves in a series of corridors and need to navigate through to get to the word tiles that form the rule “WALL IS STOP”, intending to make it more likely they will bump into the walls and experience for themselves that the walls are a barrier until



Figure 2: : Level 1 Tutorialized Version Increased Impasse Version

the rule is broken (Figure 3). Once they navigate this section, they are met with the modified second section of the level where they now see an active rule that “ROCK IS WIN”, but there is no rock object, intending to make it more likely they will face an impasse of figuring out that they must make a new rule “FLAG IS WIN”. Through the rest of the 7 initial levels, the player is faced with modifications from the original levels that increase the likelihood they will encounter impasses related to the mechanics of importance in each section of each level, giving them more tangible experience with the mechanics in the initial levels.

To investigate the influences these different modifications have on player experience, we asked individuals attending a local state fair to play one of the three versions (original, tutorialized, or increased impasse) for up to 20 minutes or until they completed levels 0-7. Participants were randomly assigned to a game version in a rolling fashion over 2 days until 75 players had participated (i.e., $n=25$ per condition). While playing, screen recordings were captured and players were recorded while vocalizing their thoughts. Gameplay logs were captured detailing the game state and all player inputs. After playing for 20 minutes or completing the initial 7 levels, each participant was asked to complete a 10-minute survey gauging their level of engagement (modified from Brockmyer, et al., 2009 [19], the term ‘engagement’ here is used as a generic indicator of game involvement. Other relevant and more technical terms

include immersion, presence, flow, psychological absorption, and dissociation), the perceived level of difficulty of their game version and each level, demographics including their general gameplay experience (a general measure of overall game familiarity created from combining “average hours of video games played per week” and “approximate number of years playing video games”), and core game mechanic content knowledge (Appendix X). For their participation, participants were offered a string backpack, in line with incentives offered by other studies in the facilities set up by the state fair.

This resulted in four streams of data: gameplay log data, screen recordings, survey responses, and talk-aloud audio recordings (talk-aloud data were outside the scope of these analyses and were not used here). Our gameplay logs were processed computationally to achieve the proper formatting and assure the fidelity of the data to participants’ actions in the game. From this dataset, we focused on log events that suggest players had made a mistake or reached an impasse: undo events (instances where players reversed one or more actions in the game), restart events (instances where players completely restarted the level), number of “no you” events (instances where players break all rules that allow control over game objects, ex. “BABA IS YOU”, making the level unwinnable), number of levels attempted, number of levels completed, number of attempts per level, and the number of times players physically encountered



Figure 3: : Level 1 Increased Impasse Version Study Design

specific barriers that demonstrate the mechanics of importance of each level. For instance, for level 1 detailed above, this was the number of times each player bumped into the surrounding walls when the rule “WALL IS STOP” was active, measuring the number of times players had tangible experience with the stop mechanic in that level.

Survey data was scored to create engagement scores, perceived difficulty scores, demographics, gameplay experience scores, and quiz scores. Quiz scores were further corrected by removing answers to questions that are based on mechanics that individual players did not play. For example, if a player did not attempt level 7, quiz answers based on the mechanics involved in level 7 were dropped for that player regardless if they were correct.

Analysis of variance tests were completed to compare the participants who played the tutorialized version, participants who played the original game, and participants who played the increased impasse version across engagement scores, perceived difficulty scores, corrected quiz scores, number of “no you” events, number of restart events, number of undo events, game play experience scores, and number of impasse encounters.

4 RESULTS

4.1 4 Demographics

Of the 75 participants that completed the study, 25 identified as male, man, “M”, or he/him; 40 identified as female, woman, girl, “F”, or cis woman; 5 identified as non-binary, agender, demigirl, or non-binary feminine; and 1 participant left gender identity blank. 4 participants completed their version of the game but did not complete the survey. The ages of participants ranged from 12 to 46 years, averaging 22 with a standard deviation of 8.

The average game experience score (general game familiarity) was 3124 with a standard deviation of 5049. However, Game experience score was much lower for participants in the tutorialized group (mean = 1415, SD = 3294) compared to participants in the increased impasse groups (mean = 3915, SD = 5737) and the original game group (mean = 3969, SD = 5481). Scores for each group contain a high level of variability with a few players reporting high game play experience while most report low or no game play experience but is markedly lower overall for participants in the tutorialized group. Given the randomized selection via rolling assignment, this difference is due to chance, but is important to consider while interpreting the results of the analyses.

Table 1: Engagement, Perceived Difficulty, and Corrected Quiz Scores

Game Version	Engagement Score	Perceived Difficulty Score	Corrected Quiz Score
	Mean (SD)	Mean (SD)	Mean (SD)
Impasse	30.79 (6.20)	26.13 (1.83)	0.81 (.14)
Original	31.29 (4.32)	25.88 (2.80)	0.81 (.12)
Tutorial	30.22 (5.24)	23.91 (2.78)	0.80 (.16)

Note. ANOVA results comparing engagement, perceived difficulty, and corrected quiz scores by group.

Table 2: Game Event Differences Among Playing Different Game Versions

Game Version	“No you” events	Undo Events	Restart Events
	Mean (SD)	Mean (SD)	Mean (SD)
Impasse	21.16 (7.10)	30.08 (17.24)	8.36 (7.31)
Original	20.80 (6.47)	24.48 (15.94)	10.16 (7.69)
Tutorial	7.88 (4.72)	16.56 (12.83)	4.56 (4.62)

Note. ANOVA results comparing the number of gameplay events by group

4.2 Group Comparisons

Average engagement, perceived difficulty, and corrected quiz scores for each group are summarized in table 1.

Analysis of variance comparing engagement scores between groups show no significant differences in overall engagement ($F(2, 72) = 0.24, p = 0.79$).

Analysis of variance comparing perceived difficulty scores between groups shows a significant difference ($F(2, 72) = 5.45, p < 0.001$). Post hoc Tukey HSD comparison reveals participants in the tutorialized version group scored significantly lower than participants in both the original version group and the increased impasse version group while no significant differences were found between the original version group and the increased impasse version group (table 1).

Analysis of variance comparing corrected quiz scores between groups shows no significant differences in overall corrected quiz scores ($F(2, 72) = 0.11, p = 0.90$).

The average number of gameplay data log events including “no you”, undo, and restart events are summarized in table 2.

Analysis of variance comparing log data events between groups shows a significant difference in “no you” events ($F(2, 72) = 37.49, p < 0.001$), undo events ($F(2, 72) = 4.84, p = 0.01$), and restart events ($F(2, 72) = 4.58, p = 0.01$). Post hoc Tukey HSD comparisons reveal that participants in the tutorialized version group produced significantly fewer “no you” events than the participants in both the original version group and the increased impasse version group while no significant differences was found between the original version group and the increased impasse version group; participants in the tutorialized version group produced significantly fewer undo events than the participants in the increased impasse version group but not the original version group, and no significant differences was found between the original version group and the increased impasse version group; and participants in the tutorialized version group produced significantly fewer restart events than the participants in the original version group, not the increased impasse version group, and no significant differences was found between

the original version group and the increased impasse version group (table 2).

Average number of levels completed, number of level attempts, and attempts per level are summarized in table 3. While we modified levels 1 to 7 for each group, these numbers include level 0, an introductory level which was not modified and all players completed at the start of play.

Average number of impasses encountered across all levels for players of the tutorialized version of the game was 14.16 with a standard deviation of 9.35, for players of the original version of the game was 25.2 with a standard deviation of 14.58, and for players of the increased impasse version of the game were 61.76 with a standard deviation of 38.00. Average number of impasses encountered across individual levels for each group are detailed in table 4. This shows that the highest proportion of impasses were encountered in the first level, especially for players of the increased impasse version, showing that the modifications to this version did, in fact, cause players to bump into the barriers of the levels more often.

5 DISCUSSION

These results shed some light on how different approaches to teaching players the mechanics of a puzzle game in the initial levels can affect their play experience. We do not see any differences between groups in their understanding of the basic mechanics of the game, regardless of the method used to impart this understanding. This may be because *Baba Is You* is already designed to encourage players to encounter impasses that demonstrate the mechanics of the game and may have already achieved a high rate of learning of the base mechanics by players already, showing a ceiling effect on learning measures compared to designing a version that increases the amount these impasses occur. Likewise, including explicit instructions about the core mechanics of the game may not have increased learning from the base game. In other words, *Baba Is You*, a very well-crafted and polished game, may have already been

Table 3: Level Statistics

Game Version	Levels complete (of 8)	Level attempts	Attempts per level
	Mean (SD)	Mean (SD)	Mean (SD)
Impasse	4.76 (1.96)	12.12 (4.19)	2.35 (.42)
Original	6.04 (1.65)	14.08 (3.75)	2.19 (.14)
Tutorial	6.84 (2.03)	15.92 (4.13)	2.13 (.22)

Note. Comparison of levels complete by group

Table 4: Impasses Encountered

Game Version	lv1 walls	lv2 left flag	lv3 water	lv4a left skulls	lv4b right skulls	lv5a no_u	lv5b lava	lv6a walls	lv6b skulls	lv6c rocks	lv6d walls destroyed	lv6e no_u wall
Impasse	33.92	8.00	2.15	1.06	4.06	1.40	1.27	0.56	0.67	5.11	8.44	1.00
Original	6.32	2.72	2.24	0.32	2.27	1.40	0.90	3.12	0.94	5.88	4.35	0.12
Tutorial	0.96	2.28	1.40	0.19	0.76	0.53	0.05	0.26	0.58	2.05	7.68	0.32

Note. Average number of impasses encountered per level by group.

designed to achieve the learning goals of the early levels at a high enough rate that modifications do not have a detectable effect.

That being said, some areas we expected to see changes in player experience for both modified versions of the game were in the player’s level of engagement and the perceived difficulty. Many times, in designing the modifications for the tutorialized version of the game, we questioned whether or not it was still fun - or even if it was still a game. However, we do not see any detectable drop in player engagement across groups, suggesting that learning the mechanics of *Baba Is You* was just as engaging through direct instruction where players were told how to complete the initial puzzles as it was to learn the mechanics through figuring it out themselves. It is important to note here that the engagement measure we used focuses on game involvement, indicating a flow-like experience as well as game enjoyment. Interestingly, the only differences we see between groups in this data are that players of the tutorialized version of the game rated their version at a lower perceived difficulty, were required to undo or restart the level less often and were more likely to complete all of the levels within 20 minutes compared to players of the original game version and players of the increased impasse version of the game. This may suggest that the most effective, albeit not necessarily best play experience for all games, is an explicit tutorial section that walks the player through the base mechanics. We do not suggest this is the best approach, as players understood the mechanics equally well in all versions and the feel of the game is also an important factor in the overall play experience, which these different approaches can greatly influence.

We do have to caveat this claim by recognizing that the players of the tutorialized version of the game incidentally scored lower on general game experience despite our conditions being randomly assigned. It could be the case that players with lower general gameplay experience might benefit from an explicit tutorial more than players with more general gameplay experience.

It is also notable that players of the increased impasse version of the game did not report a higher level of perceived difficulty despite game log data showing that they did in fact encounter more impasses via bumping into barriers, especially in the first level. This may indicate that these moments of impasse did not contribute to their sense of difficulty, suggesting that difficulty of a puzzle game lies outside of making any physical error, and more in figuring out how the mechanics work, as this was the key difference between the increased impasse and the tutorialized versions of the game. As such, complexity of the mechanics and level may be better indicators of puzzle difficulty rather than how long players spend figuring out how to progress, how many times they make mistakes, or encounter impasses. Further, contrary to conventional thinking when evaluating learning, we see here that getting stuck or making mistakes in the form of running into a barrier does not threaten learning or engagement - players can run into the walls at a much higher rate and still learn as much as being told explicitly how to complete the level, and still remain engaged in play.

A notable finding we would like to highlight is in the number of impasses that players encountered specifically in the first and second levels. Players of the increased impasse version of the game had a drastically higher average number of impasses encountered in the first level compared to players of the other versions of the game. It is true that they needed to navigate a winding corridor while players of the original game version and tutorialized version were placed in an open room, but the size of this difference is noteworthy. Especially given that in the second level, players on all three versions of the game were placed in an open room, yet the players of the increased impasse version of the game also encountered more impasses than players of the other versions. This trend also remains for nearly all of the modifications made in the increased impasse version – players interacted with the impasses more than other groups. This shows that the modifications that

were made did, in fact, increase the likelihood that players would get more tangible experience with game objects that exemplified the core mechanics. Further, the number of impasses that players of the increased impasse version of the game encountered in the second level drops dramatically and drops by half for players of the original game version but increases for players of the tutorialized version of the game. It is not clear why this increased for players of the tutorialized version of the game, but it could be that the instructions for that level were not as clear as they were in the first level. This also shows players in the increased impasse version of the game decrease in the number of times they encounter an impasse, suggesting that tangible experience with the mechanics contributes to their understanding of the mechanic, but not more than explicit direct instruction.

These findings suggest that players of each version of the game are engaged in their play and learn effectively how to navigate through the game. The only differences we see are that players of the tutorialized version of the game get through levels more quickly and easily. This suggests that the way designers build the initial levels of a game can be done so in any manner they desire with no disruption to the experience players will have. What this data does not show is how well players of these groups will perform on more challenging puzzles. Our study ends once these initial levels are complete, but this is not where players would stop playing the game. Tutorial areas and early simple puzzles are designed to prepare players for the rest of the game which will increase in difficulty and complexity. However, this study does not show how well players are prepared for further levels of *Baba Is You*. We do see players navigate through their version of the game differently, which might signal that they are learning to problem solve in different ways. This may have a long-term impact on their ability to solve new puzzles in further levels once the tutorial section of the game is complete.

In a future study, we plan to lengthen this study to show how well players of these versions of the game each do once they move past the introduction or tutorial and onto the subsequent levels of the game that do not contain any guidance to better understand the mechanics of the game. To do this, we will again ask participants to play the modified versions of the first 7 levels, complete a survey measuring their perceived level of difficulty, engagement, enjoyment, and content knowledge. Afterwards, they will all be asked to play through the same original game levels in the first sub-world, The Lake, that follows the first 7. This extension will show how these different tutorial design styles affect players' long-term progression.

6 CONCLUSIONS

This study compared two modified versions of the popular puzzle game *Baba Is You* to investigate if two popular styles of teaching the player about the mechanics of the game, direct instruction through an explicit tutorial or productive failure through an implicit tutorial, showed any differences in how well players learn, how engaged they are, and how difficult the game was for them. We recruited participants at a local fair to play one of the modified versions or the original game for up to 20 minutes or until they completed the first 7 levels of the game. Once complete, participants were

asked to complete a 10-minute survey gauging their engagement (game involvement and enjoyment), their perceived difficulty, and their content knowledge of the game's mechanics introduced in the first 7 levels. Our analyses show that players of the tutorialized version of the game had an easier time completing the first 7 levels, seen through lower rated levels of difficulty as well as encountering fewer impasses that resulted in loss of player control, restarting the level, or undoing previous actions compared to players of the other versions. Meanwhile, players of all versions scored equally well on the content knowledge assessment, suggesting that the different approaches to teaching the mechanics of the game did not have any influence on how well the player learned how those mechanics worked. While this does suggest making a more explicit tutorial will decrease the difficulty players experience, it does not suggest that they will learn the mechanics better or be less engaged in the game. These results give some indication that tailoring the introductory levels to the desired difficulty feeling of the game either towards an implicit or explicit style tutorial won't cause players engagement levels or understanding of the game mechanics to drop. However, it remains unclear how well-prepared each of these versions leave players after the first 7 levels. Arguably, since players of each version scored the same on the content knowledge assessment, they would each be equally prepared to progress through the game. However, the learning styles implemented in these changes may leave some players with better meta strategies that enable progression in difficult levels that require a more purposeful approach and higher effort in levels beyond the early tutorial levels. To address this, we plan to expand this work to investigate how well players of these modified levels progress through more difficult levels after learning the basic mechanics.

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